



RESEARCH ARTICLE

Detection and localization of pipeline leakage of water distribution networks based on graph convolutional networks

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Abstract

A leak in a water distribution network (WDN) creates a localized pressure depression whose effect diffuses along the network topology through head-loss coupling. This spatial diffusion is well approximated by graph Laplacian smoothing, so we use a graph convolutional network as a learnable surrogate of hydraulic information propagation on the graph. The method casts single-snapshot leakage analysis as graph-level classification of normal versus abnormal states using topology-aware convolution. Localization then leverages gradient-based node importance—sensitivities of the leak logit to node features—mapped to pipes via linear interpolation to produce leakage-risk contour maps. The framework couples pressure and demand features, accommodates partial monitoring by varying sensor density, and evaluates noise robustness by injecting Gaussian noise into inputs. In simulated WDNs, the approach maintains high accuracy across sensor-density settings and exhibits a predictable trend under increasing noise, where false negative rate rises faster than false positive rate as signal-to-noise ratio decreases. On the L-Town benchmark, we use real pressure measurements and nodal demand data consistent with metered consumption (generally <10% deviation), providing a realistic testbed. A simplified network representation accelerates convergence while preserving localization fidelity, enabling millisecond-level inference suitable for operational deployment. Together, these results support a physically grounded, computationally efficient pathway for leak detection and localization in smart water networks.

1 | INTRODUCTION

The water distribution network (WDN) is vital to urban infrastructure, directly impacting urban sustainability (Nayak & Turnquist, 2016). With rapid urbanization, WDN operational loads and leakage frequencies have risen sig-

nificantly (Adedeji et al., 2017). Studies estimate that global WDN leakage costs reach \$39 billion annually (Daniel et al., 2022). Timely leakage detection and effective management are thus essential to safeguard urban water supplies and minimize economic losses (Fan et al., 2022).

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Detection and localization of pipeline leakage play a critical role in maintaining water supply safety. Real-time monitoring and data analysis enable timely identification of potential leakages risks, allowing for effective repair measures (Garlisi et al., 2022). However, existing monitoring and detection methods still face limitations in accuracy and timeliness (Luo et al., 2024). Traditional methods often struggle to meet real-time detection requirements, especially in complex urban network environments (Kakeshpour et al., 2024).

Early applications of deep learning in civil engineering demonstrated its capability to model nonlinear and high-dimensional problems across different subfields.

Deep neural architectures were first employed to estimate real estate prices by integrating socioeconomic indicators through hybrid learning frameworks, showing strong predictive performance for large, complex datasets (Rafiei & Adeli, 2016). Similar approaches were later applied to predict the compressive strength of concrete mixtures, where deep restricted Boltzmann machines achieved higher accuracy and computational efficiency compared with traditional regression and shallow neural networks (Rafiei et al., 2017). In the area of disaster prevention, neural dynamic models were developed for earthquake early warning, combining machine learning classification and optimization algorithms to forecast earthquake magnitude and location (Rafiei & Adeli, 2017a). These early studies collectively revealed the potential of deep learning to capture nonlinear and temporal dependencies in civil engineering systems, paving the way for its adoption in WDN analysis and other intelligent infrastructure applications.

In the current study, both fully supervised and semi-supervised methods for analyzing Supervisory Control and Data Acquisition (SCADA) monitoring data of WDN have been widely explored. In terms of fully supervised learning, Peng et al. (2022) proposed a burst pipe region identification model based on a fully connected network, FCN-DenseNet, and optimized the hyperparameters using Bayesian optimization, significantly improving the model's identification accuracy, reaching 93.5% and 96.7%. An intelligent system was presented for predicting incident reports (IRs) in sectorized WDN, such as drains in sidewalks, lack of pressure, lack of water, leaks, or others, based on pressure and flow data (Robles-Velasco et al., 2025). Liu et al. (2024) improved the extreme learning machine (GELM) algorithm using a Gaussian mixture model (GMM), overcoming traditional ELM's limitations in classification accuracy and significantly improving leakage identification accuracy. Antunes et al. (2023) optimized the hyperparameters of the convolutional neural network (CNN) to enhance burst pipe localization accuracy, showing good adaptability under different noise levels and burst

pipe positions. Fan and Yu (2022) constructed a framework combining probabilistic long short-term memory (LSTM) networks with Monte Carlo methods for uncertainty quantification in pipeline leakage rate prediction across the entire system. The results showed that its prediction accuracy outperformed traditional ARIMAX models and effectively narrowed the uncertainty range. Cai et al. (2025) coupled a real-time transient mechanistic model derived from the Bernoulli equation with statistical and regression learning, fitting real-time features using normal-operation data and triggering early warnings via probability-of-detection and probability-of-false-call thresholds, thereby maintaining a high detection rate even with limited leakage samples.

In terms of semi-supervised learning, Tornyeviadzi and Seidu (2023) proposed a one-dimensional (1D) CNN deep autoencoder, which was trained in a semi-supervised manner to detect and locate leakage using multivariate time-series data, efficiently identifying and localizing leaks without the aid of hydraulic models. Karimanzira (2023) developed a hybrid autoencoder model with a 3D CNN encoder and ConvLSTM decoder, combined with a spatiotemporal attention mechanism, which enhanced the robustness of leakage detection and localization accuracy, successfully improving leakage detection accuracy to 96% while reducing the false positive rate (FPR) to 4%. Under a gas–water two-phase transient environment, Zhang et al. generated samples using a discrete gas cavity model for water-hammer simulation and performed localization with supervised learners such as a genetic-algorithm–backpropagation network; even in high-noise scenarios, the approach achieved percent-level localization error (MAPE $\approx 9.26\%$), demonstrating the applicability of deep supervised models under complex operating conditions (Q. L. Zhang et al., 2024).

In addition, in contrast to fully supervised and semi-supervised approaches, analytical methods emphasize reproducible, derivable closed- or semi-closed-form solutions grounded in the 1D unsteady-flow governing equations. For example, transfer-matrix and frequency-response-function formulations have been used to reconstruct transient–leakage coupling, where multi-sensor spectral information is fused to suppress sidelobes and enhance the resolvability of multiple leaks (Li & Zhang, 2024). Other researchers have derived Fourier-series analytical solutions for cracks and distributed outflows. These solutions were then validated against the method of characteristics; the results reveal how crack length and leakage ratio drive water-hammer peak attenuation and resonance-frequency drift (Y. Zhang et al., 2024).

Although these studies have made significant progress in data-driven modeling, time-series prediction, anomaly



detection, model interpretability, and model optimization, they are all fundamentally based on Euclidean space (Montáns et al., 2019).

In Euclidean space, signal processing and perception are relatively straightforward, and data can typically be embedded into a regular vector space. However, the connections between nodes and pipes in WDN are complex, involving not only distances and the distribution of sensor locations but also irregular network topologies (Liang et al., 2021). Such network topologies often cannot be easily represented in low-dimensional Euclidean space, making it difficult for traditional time-series models to fully capture the relationships and dependencies among WDN nodes. Signal perception and analysis in non-Euclidean space are more complex compared to traditional Euclidean space. To effectively incorporate topological relationships, models must adapt to non-Euclidean data processing. Previous research has demonstrated the adaptability of graph theory to non-Euclidean domains by introducing general complexity measures for weighted and irregular networks, extending beyond small-world or scale-free assumptions (Ahmadlou & Adeli, 2017). Such models can map the structural information of WDN (nodes, edges, and their attributes) into a graph feature space, using convolution mechanisms to uncover potential spatial associations within the network structure (Miao et al., 2023).

In recent years, graph convolutional networks (GCNs) have gained attention for their advantages in processing graph-structured data (Fan et al., 2022). Using GCN for WDN modeling can effectively utilize the network topology, improving detection accuracy and efficiency (Salem et al., 2024). Through graph convolution operations, GCN can propagate information between nodes, enabling comprehensive analysis of the network's status (Fan & Yu, 2022). They employed a Graph Convolution Network–Deep Reinforcement Learning (GCN-DRL) model to determine the optimal repair sequence for a testbed WDN subjected to earthquake-induced damage. Sahin and Yuce (2023) proposed a leakage-prediction framework based on GCN, which was driven by hydraulic time series collected on an experimental rig under both normal and leaking conditions and achieved an accuracy of 94%. The rig comprised eight pressure sensors (10 Hz), two flow meters, and PVC pipelines from which the hydraulic data were obtained for both normal and leaking states. However, it was not validated on an operational network. Li and Wu (2024) developed a GCN based on a segment feature fusion strategy, driven by current pressure, predicted pressure, and pressure fluctuation features. The model was validated on the WDN of City H using hydraulic data computed with EPANET 2.2. Results showed that, with a pressure-sensor coverage of 70.9%, the algorithm identified 96.1% of leaks whose topological distance was 3

(graph hops). Only serialized pressure signals were used to drive the model; water-demand measurements were not considered. Robles-Velasco et al. (2025) introduced a deep-learning methodology for incident detection in sectorized WDN. The approach used real sensor data and was validated on a city-scale WDN in Seville, Spain. However, the study did not examine the implications of adopting a GCN framework.

Longer feature-input sequences typically imply that these models require greater computational resources. In fact, the aforementioned GCN-based studies did not report computational costs, which hampers a practical assessment of their deployability in engineering settings. Moreover, the sampling frequency of monitoring data in WDN often falls short of these algorithms' requirements. Therefore, it is necessary to investigate GCNs driven by single-time-snapshot hydraulic information for leakage identification. In real WDN, a primary challenge lies in appropriately embedding network topology and associated attributes into the model. Second, many current GCN approaches assume relatively dense sensing, whereas practical deployments are sparsely instrumented, in sharp contrast to the data richness typically expected by GCN. Accordingly, progressively reducing the number of monitored nodes used by the GCN to determine the minimum sensor set for reliable leakage identification becomes crucial; this can preserve accuracy while lowering monitoring costs and improving the feasibility of GCN-based leakage detection and localization in WDNs. Finally, the joint dynamics of nodal demand and pressure in WDNs can help distinguish legitimate consumption events from leakages: an increase in demand accompanied by a pressure drop typically indicates normal operation; conversely, a pressure decrease under steady or declining demand often signals a leakage. Hence, it is necessary to jointly incorporate demand and pressure measurements into the model and algorithmically exploit their synergy.

To enable rapid leakage detection and localization, a GCN driven by single-time-snapshot hydraulic states of the WDN was developed. Leveraging topology-aware aggregation, the model jointly encodes the relationships among monitored nodal pressures, nodal demands, and leakage states. The workflow first performs supervised whole-graph classification on the embedded WDN graph, and then uses nodal importance obtained during classification to infer leaking pipe locations. The effect of sensor coverage was investigated by validating the approach on a simulated WDN and analyzing how the number of monitoring points influences robustness. For real-world assessment, the method is further applied to the L-Town WDN. The network representation is extracted and simplified using pipe flow direction and nodal-pressure characteristics, merging nodes similar to instrumented pressure

nodes into their corresponding monitoring nodes to better align the algorithm with operational system models. The results demonstrate that the proposed method effectively identifies leaking pipe segments.

2 | METHODOLOGY

Similar to GCN, WDN modeling also relies on graph-structured data. WDN can be naturally represented as a graph, where nodes represent water usage points, and edges represent pipeline connections. This structure perfectly aligns with the capabilities of GCN in processing graph data, enabling effective coupling for efficient modeling and analysis.

In WDN modeling, nodes typically have various feature information, such as pressure, flow, and water demand. GCN, through its convolutional mechanism, aggregates the information of a node and its neighboring nodes to generate a new feature representation for the node. In this way, each node in the WDN not only depends on its own features but also leverages the information from neighboring nodes for modeling. For example, the pressure of a node might be influenced by the flow in adjacent pipelines or the pressure at neighboring nodes. GCN effectively combines this neighboring information to provide a more accurate feature representation.

GCN typically consists of multiple layers, where each layer aggregates features from nodes and their neighbors, progressively expanding the node's receptive field. In WDN, this means that the model can gradually learn more global graph information from the local topology, allowing it to better simulate the propagation and distribution of water flow in the complex network. Multi-layer GCNs can capture deeper network dynamics, including relationships between distant nodes. These similar characteristics suggest that using GCN for WDN modeling may yield promising results.

This study is based on the connectivity information (topology), node information (water usage and pressure), and state information (graph labels) of the WDN. By embedding the WDN information into a graph, it achieves the modeling of the WDN and the identification and localization of pipeline leakage. First, the coupling mechanism between the WDN and GCN is validated by simulating the precise node information within the WDN. After successful validation, the number of input features is gradually reduced to explore the minimum number of monitoring points required to meet the algorithm's requirements. Next, the normal and abnormal state from the L-Town WDN are used to assess the applicability of this algorithm in large-scale, real-world WDN.

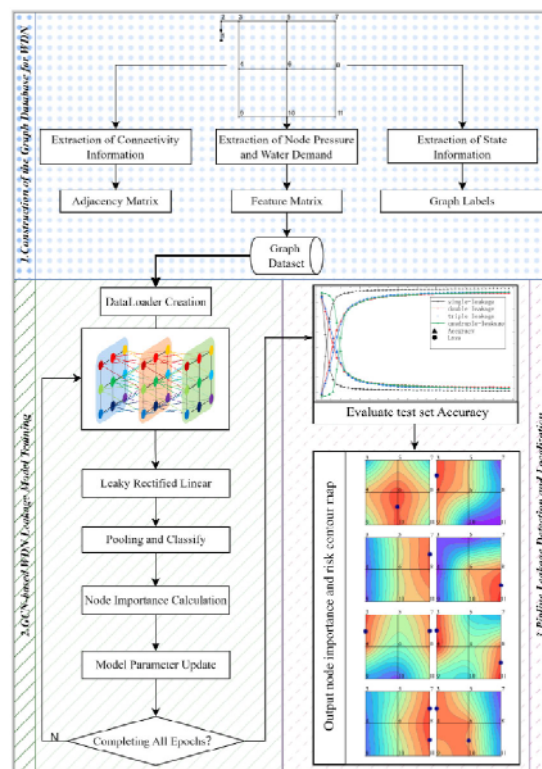


FIGURE 1 Leakage detection and localization framework. (1) Construction of the Graph Database for WDN; (2) GCN-based WDN Leakage Model Training; (3) Pipeline Leakage Detection and Localization.

2.1 | Leakage detection and localization framework of WDN based on GCN

GCN is a powerful tool for processing graph-structured data (which will be specifically introduced in Section 2.2). Since its underlying logic is based on irregular graph data, this algorithm can be effectively coupled with WDN modeling. In this study, a leakage detection and localization framework (Figure 1) was developed, consisting of three main components: construction of the graph database (Section 2.3), model training (Section 2.4), and output (Section 2.5).

First, the connectivity, node information, and graph states (whether leakage occurs or not) in the WDN are integrated into a graph, forming a graph database. Each graph in this database represents the single-time-snapshot information of the WDN under the corresponding state. Then, iterative classification training is performed on the model, which reflects the process of aggregating water-related information along the topology of the WDN and the process of making classification decisions. Finally, the importance of nodes in the decision-making process is used to locate pipeline leaks.

2.2 | Principles of graph convolutional network

GCN is a class of deep-learning models specifically designed for processing graph-structured data. Traditional CNNs, which excel at handling grid-like data structures such as images or sequential data, perform relatively poorly when applied to non-Euclidean data structures. These non-Euclidean data types include graph-structured data, such as transportation networks, molecular structures, and WDN (Derr et al., 2018).

To address the challenges of processing non-Euclidean data, GCN was proposed. It extends the convolution operation from regular grid data to graph data, enabling effective extraction of features and structural information from both the nodes and their neighbors in the graph (Li et al., 2018). GCN aggregates and updates information for each node in the graph, allowing the node's features to include both its own and its neighbors' information.

A typical GCN model consists of multiple convolutional layers, where each layer progressively processes the node features. In each layer, each node updates and aggregates its features based on the information from its neighbors. Ultimately, the output after multiple convolutional operations reflects the global structural characteristics of the nodes in the graph. This model can be widely applied to various tasks, such as node classification, graph classification, and link prediction (Niepert et al., 2018).

2.3 | Construction of the graph database for WDN

The GCN algorithm uses a graph database as input, thus the data in the WDN needs to be organized into a graph database format that can be recognized by GCN to extract the relevant information. This study proposes a method for organizing WDN data into a graph database format, which consists of the network's connectivity information (topology), node information (water consumption and pressure), and state information (graph labels). The GCN captures the relationships between the network's connectivity, node information, and state information, enabling the classification of the network's state.

A graph is typically represented as $G = (V, E)$, where V denotes the nodes and E denotes the edges. In the graph representation of a WDN, the water-demand junctions, pressure monitoring points and pipe junctions can be considered as the nodes V in the graph, while the pipes can be considered as the edges E . The connection relationships of nodes through the pipes form an $N \times N$ adjacency matrix A (where N is the number of nodes; for the example network

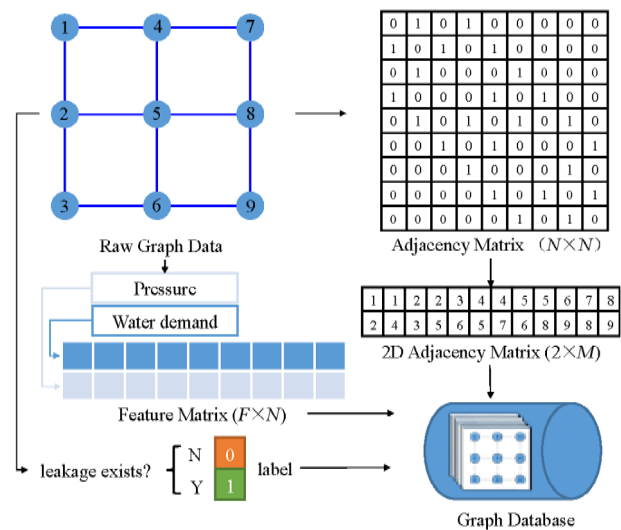


FIGURE 2 Graph database construction.

in Figure 2, $N = 9$). To simplify storage in WDN modeling, the adjacency matrix is reduced to a $2 \times M$ adjacency matrix (where M is the number of pipes; for the example network in Figure 2, $M = 12$). The demand for water at the water-demand junctions and the pressure at the pressure monitoring junctions are used as features and placed into the node-feature matrix ($F \times N$), where F is the number of features per node; in this study $F = 2$. Whether leakage exists in the WDN is included as a label in the graph dataset. The label is defined as a binary variable: 0 indicates that no leakage is present in the network at that time, and 1 indicates that a leakage is present. These labels are used both to train the GCN (i.e., to specify whether each graph instance should be classified as “leakage” or “normal”) and to evaluate the trained model on the test set (i.e., to verify whether the learned decision rule is correct). The overall process of graphing the WDN information is shown in Figure 2.

2.4 | GCN-based WDN leakage model training

In GCN, information is aggregated from each node and its neighboring nodes through convolution operations, allowing the representation of each node to incorporate the features of its neighbors, thereby capturing the local structure of the graph. The core of this process is to apply a linear transformation at each node and aggregate the information from its neighbors through graph convolution. In WDN, this mechanism of GCN effectively aggregates water-related node data, such as pressure and water usage, along the network's topology. Through this aggregation,



the model captures the propagation relationships of water flow and pressure across different pipes, as well as the dependencies between monitoring points.

2.4.1 | Training computation process for GCN

In this study, the optimization of the pure GCN convolution kernel is now being emphasized, with it being effectively integrated with the practical challenges of leakage analysis. As shown in Equation (2), information is aggregated by the GCN model from each node and its neighbors, and deeper graph structural information is progressively extracted layer by layer.

From the standpoint of leakage-induced hydraulic perturbations, nodal pressures in a WDN are lower under leakage than under no-leak conditions when demand is held constant, and different leakage locations produce distinct combinations of pressure-drop magnitudes at monitoring points. Without demand measurements, such pressure-drop patterns cannot be reliably attributed to leakage versus legitimate consumption. Accordingly, the joint pressure–demand signature is used to drive the GCN.

Node-level hydraulic features (pressure and demand) are propagated through the GCN's convolutional kernels, whereby node information is aggregated. Physically, leakage occurrence is inferred from the joint behavior of nodal pressures and demands, and the leakage location is determined by interrogating the model's decision process via nodal importance during classification.

The final design and training process of the GCN-based WDN leakage model is presented below:

- A. Data Input: The graph data (including the adjacency matrix, feature matrix, and graph labels) is loaded into the data loader, with real hydraulic parameters from the WDN being directly related to potential leakages.
- B. First Convolution (GCNConv1): The input features are mapped from the original dimension to a hidden dimension, with the matrix $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)}$ being employed to blend the water-demand and pressure information from neighboring nodes, thereby initiating the propagation of hydraulic states along the network topology and integrating local leakage clues into the analysis.
- C. Nonlinear Transformation: The leaky rectified linear unit (Leaky ReLU) activation function is employed, enhancing the model's sensitivity to subtle variations in the hydraulic signals, which may serve as indicators of leakage.
- D. Second Convolution (GCNConv2): The features from the hidden dimension are mapped to the output dimension, further refining the propagation of

hydraulic information and making the leakage-related patterns more distinct, while the convolution process is dynamically adjusted by introducing a learnable weight matrix $W^{(1)}$ based on classification accuracy.

- E. Pooling: The node features are aggregated into a comprehensive graph-level representation, a step that captures the overall hydraulic state of the network and is regarded as crucial for identifying anomalies caused by leakages.
- F. Loss Computation: The cross-entropy loss(CE loss) between the model output and the ground truth labels, which represent verified leakage conditions, is computed. When a misclassification occurs and the model assigns high confidence (p) to the incorrect class, the cross-entropy loss becomes large. Its definition is given in Equation (1).

$$L = -[y \log p + (1 - y) \log (1 - p)] \quad (1)$$

where L is CE loss for a single sample(smaller values indicate better performance) and y is the ground-truth label of the sample (0 denotes normal, 1 denotes leakage).

- G. Gradient Computation: The gradients of the loss with respect to the model parameters are calculated.
- H. Gradient Norm Check: The global norm of all parameter gradients is computed to ensure that it does not exceed a predefined maximum value, thereby maintaining training stability, which is used to prevent overly large gradients from skipping over the optimal leakage features of the WDN.
- I. Parameter Update: The Adam optimizer is employed to update the model parameters using the computed gradients, thereby further refining the model's ability to detect leakage-related patterns in the WDN.
- J. Loss Averaging: The average CE loss for the current epoch is calculated, providing a quantitative measure of the model's overall performance in capturing leakage anomalies in the WDN.
- K. Convergence Check: If the CE loss falls below 0.01 or the maximum number of epochs is reached, the training loop is terminated and the final model is output; otherwise, steps B through K are repeated, ensuring that the model sufficiently learns the features required for leakage detection in the WDN.

The convolution operation can be simplified as Equation (2):

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}) \quad (2)$$

where, $H^{(l)}$ is the node-feature matrix of the l th layer; $\tilde{A}$ is the adjacency matrix with self-connections; $\tilde{D}$ is the degree



matrix of $\tilde{A}$; $W^{(l)}$ is the learnable weight matrix; σ is the nonlinear activation function.

2.4.2 | Parameter settings of GCN

In this study, Leaky ReLU is introduced to avoid the vanishing gradient problem and improve the model's generalization ability. Leaky ReLU is employed with a negative slope set to 0.01 and the inplace flag set to false. It is a commonly used activation function in deep learning (Xu et al., 2020), and represents an improvement over the standard rectified linear unit (ReLU). Cross-entropy loss, which originates from information theory and is used to measure the difference between two probability distributions, is utilized for classification tasks (Connor et al., 2024). Specifically, it quantifies the additional amount of information required to express the true distribution $f(y)$ using the model's predicted distribution $f(p)$ and produces a larger CE loss for high-confidence incorrect predictions to encourage adjustment. In this study, the cross-entropy loss is configured with the reduction mode set to "mean," an optional weight parameter to address class imbalance, and an ignore index parameter for selectively excluding specific classes from CE loss computation. The Adam optimizer is used as an adaptive learning rate optimization algorithm that combines momentum and adaptive gradient methods (Li et al., 2024). It adjusts the learning rate for each parameter by computing the exponentially weighted moving averages of the first and second moments of the gradients, thereby enhancing training efficiency and stability. In this study, the Adam optimizer is configured with a learning rate (lr) of 0.001, beta parameters (betas) set to (0.9, 0.999), epsilon (eps) set to 10^{-8} , and an optional weight decay parameter for regularization.

2.5 | Pipeline leakage detection and localization

The model's output is divided into two parts. The first part is the mapping relationship between the graph labels, graph topology, and node information. This relationship can be used to classify the state of the network. In other words, the graph topology and node information are input into the mapping relationship to identify whether the network is in a normal or abnormal state. The second part of the model's output is a leakage area warning, which identifies leakage regions based on the importance of nodes in classification.

2.5.1 | Pipeline leakage detection

The GCN model aggregates node features and connectivity in the WDN to learn representations capturing hydraulic variations. After training, it distinguishes normal operations from leakage-induced anomalies. Its output is converted into class indices using $\text{argmax}(\text{dim} = 1)$, resulting in the predicted labels for each graph sample.

The predicted labels are compared with the true labels to calculate the number of correct predictions and the total number of samples. The final identification accuracy is the ratio of the number of correct predictions to the total number of samples. Its definition is given by Equation (3):

$$\text{ACC} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{y}_i = y_i\} \quad (3)$$

$\hat{y}_i$ is the predicted label for the i th sample, y_i is its true label, and N is the total number of samples. $\mathbf{1}\{\hat{y}_i = y_i\}$ is an indicator function, which returns 1 if the predicted label matches the true label, and 0 otherwise.

To further explore the identification details, Equations (4) and (5) for the false negative rate (FNR) and FPR are defined.

$$\text{FNR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{y}_i = 0 \mid \hat{y}_i \neq y_i\} \quad (4)$$

$$\text{FPR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{y}_i = 1 \mid \hat{y}_i \neq y_i\} \quad (5)$$

2.5.2 | Pipeline leakage localization

In a WDN-oriented GCN, hydraulic couplings between nodes are modeled as information propagation over the graph (Equation 2). The convolution operator can be viewed as a smoothed form of the graph Laplacian, which performs weighted aggregation of node features under topological constraints. As a result, perturbations of node attributes propagate along edges and affect the global output $o = [o_{\text{normal}}, o_{\text{leak}}]$. By backpropagating through the leakage-related output component, node-wise sensitivities to the leakage response are obtained. Backpropagation is defined in Equation (6).

$$\frac{\partial o}{\partial X} = \frac{\partial o}{\partial H^{(2)}} \cdot \frac{\partial H^{(2)}}{\partial H^{(1)}} \cdot \frac{\partial H^{(1)}}{\partial X} \quad (6)$$



where X denotes the set of features for all nodes, $H^{(1)}$ denotes the node-feature matrix produced by the first convolutional layer, and $H^{(2)}$ denotes the node-feature matrix produced by the second convolutional layer.

The importance of node i is defined in Equation (7).

$$S_i = \sum_k \left| \frac{\partial o}{\partial x_{i,k}} \right| \quad (7)$$

where S_i is the importance of node i ; $x_{i,k}$ is the k th feature of node i ; and $\frac{\partial o}{\partial x_{i,k}}$ is the gradient of the output o with respect to the k th feature of node i .

This gradient, obtained via backpropagation, quantifies how local perturbations of node features—after propagating through the graph—affect the leakage-related output. Since message passing in GCNs is mathematically equivalent to Laplacian smoothing on the adjacency structure, $\frac{\partial o}{\partial x_{i,k}}$ can be regarded as a sensitivity propagation coefficient for leakage response diffusing along edges; accordingly, the node importance characterizes both the spatial extent and the magnitude of a local hydraulic disturbance's impact on the overall system.

Because leakage occurs on pipe segments rather than at nodes, node-level importance is mapped to edges. From a hydraulic perspective, pressure varies continuously along a pipe and, under small perturbations, can be linearized using the Hazen–Williams relation, so inter-nodal responses are treated as linearly transmitted along the segment. To maintain spatial continuity, linear interpolation is applied along each pipe's length (Equation 8)

$$S_{ij}(\tau) = (1 - \tau)S_i + \tau S_j, \tau \in \left[0, \frac{1}{6}, \dots, 1\right] \quad (8)$$

Accordingly, node importance can be derived for both pipe segments and nodes in the network graph. In this study, node importance was employed to indicate the locations with the highest likelihood of leakage (leakage risk) when a sample is classified as the leakage state. By combining the magnitudes of node importance with nodal positions, a leakage-risk contour map of the WDN can be constructed.

To further evaluate the localization accuracy of the model in the leakage localization task, the two indicators: T (Top- r accuracy) and T' (Top- $r + 1$ accuracy) were introduced, which are widely used in multi-classification problems to assess the model's localization performance. In this study, these two indicators are used to evaluate the localization accuracy when there are n leaking pipes in the WDN.

T refers to the proportion of cases where the r leaking pipes localized by the model exactly match the actual leaking pipes. Its definition is given by Equation (9).

$$T = \frac{1}{R} \sum_{j=1}^R \mathbf{1} \left\{ P_{1:r}^{(j)} = A^j \right\} \quad (9)$$

where R is the total number of test cases; $P_{1:r}^{(j)}$ is the set of the top n pipes localized by the model in the j th case; A^j is the actual set of leaking pipes in the j th case; and $\mathbf{1}\{\}$ is an indicator function that equals 1 if the condition inside is true, and 0 otherwise. T' indicates the proportion of cases where all leaking pipes are contained within the first $r + 1$ pipes localized by the model. This indicator is relatively lenient, reflecting that even if the model does not rank the correct risk level as the top one, it still places it among the top positions. Its definition is given by Equation (10).

$$T' = \frac{1}{R} \sum_{j=1}^R \mathbf{1} \left\{ P_{1:r+1}^{(j)} \supset A^j \right\} \quad (10)$$

where

R is the total number of test cases; $P_{1:r+1}^{(j)}$ is the set of the top $n + 1$ pipes localized by the model in the j th case; A^j is the actual set of leaking pipes in the j th case; $\mathbf{1}\{\}$ is an indicator function that equals 1 if the condition inside is true, and 0 otherwise.

By introducing the T and T' indicators, not only can the model's optimal localization capability be strictly measured, but the overall grasp of the risk levels by the model can also be understood through its suboptimal localization outcomes. If T' is significantly higher than T , it indicates that while the model demonstrates a certain sensitivity in recognizing the correct risk level, there is still room for improvement in the final decision-making, which provides strong evidence for subsequent model tuning and decision mechanism optimization.

3 | RESULTS AND DISCUSSION

3.1 | Validation experiments of a simulating WDN

First, this study used a computer-simulated WDN model along with corresponding network state information (including leakage pipe locations, node pressures, and node demands). The simulation was conducted using EPANET and water network tool for resilience (WNTR), allowing for better batch processing, and provided a

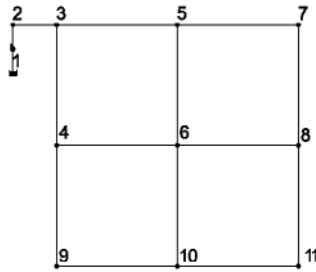


FIGURE 3 Schematic diagram of the simulated water distribution network.

theoretical validation of GCN for leakage detection in the WDN. The experimental validation was organized into three stages: (1) full-monitoring validation, (2) partial-monitoring validation, and (3) robustness assessment. In the full-monitoring stage, accurate pressures and demands from nine consumer nodes were fully provided to the model input to test the hypothesis that hydraulic information can be propagated through graph convolution layers of a GNN. In the partial-monitoring stage, the inputs comprised only a subset of nodal measurements: demand data were available for 44.4% of all nodes and were intentionally perturbed to yield a maximum relative error of 20% with respect to the ground truth; pressure data were supplied at coverage levels of 100%, 44%, 22%, and 11% to examine the algorithm’s sensitivity to leakage events under progressively reduced monitoring density. During the robustness assessment, input noise of varying intensities was injected to evaluate robustness. Validation under practical conditions using the L-Town network—where both demand and pressure include real measurement noise—will be presented in Section 3.2.

3.1.1 | Introduction of a simulating WDN model

The WDN model used for the validation experiments is shown in Figure 3. As illustrated in Figure 3, nodes are represented by numbers. The entire simulated WDN consists of 1 water source (Node 1), 10 consumption nodes (Nodes 2 to 11), 4 leakage points, 17 pipe, and 1 water pump.

Four leakage points were respectively added to four pipes (pipes 3–4, pipes 7–8, pipes 6–10, and pipes 8–11). These combinations are used as leakage pipe inputs in EPANET, enabling the simulation of leakage conditions in real-world WDN within EPANET.

The simulation is divided into two parts: normal conditions and leakage conditions.

a. Normal Conditions:

Pressure-dependent demand (DDP) model is adopted to calculate the hydraulic state of the simulated WDN. Water consumption data are sourced from the monitoring database of a WDN in a city in Northern China. These measured data are integrated into the water consumption nodes of the simulated WDN for hydraulic calculations, with a time step of 1 min. Consequently, the node pressure data under normal WDN conditions can be obtained.

b. Leakage Conditions:

Based on the normal conditions, leakage points are introduced into the simulated WDN to represent abnormal leakage scenarios. Because leakage discharge follows a power-law relationship with pipe pressure, the leakage variables in the model are updated in real time. The leakage rate is approximated by the classical orifice outflow relation from fluid mechanics. The method for calculating the leakage volume is given in Equation (11).

$$Q = C_d A \sqrt{2gH} \quad (11)$$

Q is the leakage flow rate (L/s); C_d is the discharge coefficient for the orifice (0.6); A is the orifice area (m²); g is the gravitational acceleration (9.81 m/s²); and H is the pressure (m).

In EPANET, the node-level emitter coefficient characterizes the outflow capacity of a node, namely the discharge that can be expelled from the system under a specified pressure. The empirical relation is provided in Equation (12).

$$Q = C_e P^{0.5} \quad (12)$$

where C_e is the emitter coefficient ($L/s/m^{0.5}$); and P is the pressure (m).

By comparing Equations (11) and (12), it can be found that $C_e = C_d A \sqrt{2g}$. Circular-orifice leaks with different areas can be embedded into the EPANET model. To reflect different leakage severities, distinct orifice diameters are assigned: 5 mm for pipes 3–4, 10 mm for pipes 7–8, 15 mm for pipes 6–10, and 20 mm for pipes 8–11. The steady-state characteristics of the WDN induced by leakage were taken as the basis of this study, and emphasis was placed on the differences in nodal pressures and demands between leakage-present and leakage-absent states. Within this scope, pipe material, soil conditions, aging, and crack-propagation mechanisms were regarded as having only a minor influence. Consequently, leakage was modeled in a simplified manner as circular-orifice discharge.

Leakage scenarios are categorized into four types: single leakage, double leakage, triple leakage, and quadruple leakage. Taking single leakage as an example, independent

leakages are simulated individually on pipes 3–4, 7–8, 6–10, and 8–11, totally $4(c_4^1)$ models. Similarly, the double-leakage, triple-leakage, and quadruple-leakage scenarios generate $6(c_4^2)$, $4(c_4^3)$, and $1(c_4^4)$ model(s), respectively, resulting in a total of 16 distinct leakage condition models.

Since GCN require a large number of samples as input data, the simulation calculations are conducted using WNTR. WNTR is an open-source hydraulic analysis tool that supports efficient simulation and batch processing for WDN. By employing WNTR, researchers can deeply analyze network dynamics, store simulation results, and provide foundational data for subsequent GCN modeling. A batch single-time-snapshot hydraulic state simulation algorithm based on WNTR is proposed to rapidly obtain the hydraulic states under various operating conditions. The basic computational logic is shown in Figure 4.

The algorithm dynamically updates node demands in the WDN model and conducts large-scale hydraulic analysis through stepwise simulations. First, necessary libraries are imported, and the WDN model (.inp) and water-demand data (CSV) are loaded. Then, demands at nodes are iteratively updated according to each row of data, hydraulic simulations are executed using the WNTR-Simulator, and node pressures are retrieved. Results are stored in a pandas DataFrame and written to an output file in batches to reduce I/O operations. After each simulation, the WDN model state is reset to ensure independent demand conditions for subsequent simulations. Upon completion, the CSV file is closed to ensure data integrity. The algorithm thus efficiently performs large-scale WDN hydraulic simulations through batch processing, dynamic demand updates, and optimized file handling.

In the hydraulic state calculation for the simulated WDN, a total of 14,400 sets of hydraulic states under normal conditions were calculated (with a data resolution of 1 min and a total duration of 10 days). For leakage conditions, simulations were performed for 16 distinct leakage scenarios, with each scenario containing 14,400 sets of single-time-snapshot hydraulic states, resulting in a total of 230,400 abnormal single-time-snapshot hydraulic state samples. Accurate demand and pressure measurements are used to populate graph instances for the full-feature model analysis (Section 3.1.2). For the sensor-coverage study (Section 3.1.3), accurately computed pressures are assigned as nodal pressure features in the graph database, while demand features are formed by adding 20% Gaussian noise to the demand data. The leakage state (1 for presence; 0 for absence) is stored as the graph label and is employed for subsequent training and the evaluation of validation metrics. To facilitate subsequent simulation processes, the data were grouped into four categories based on leakage conditions: single-leakage database, double-

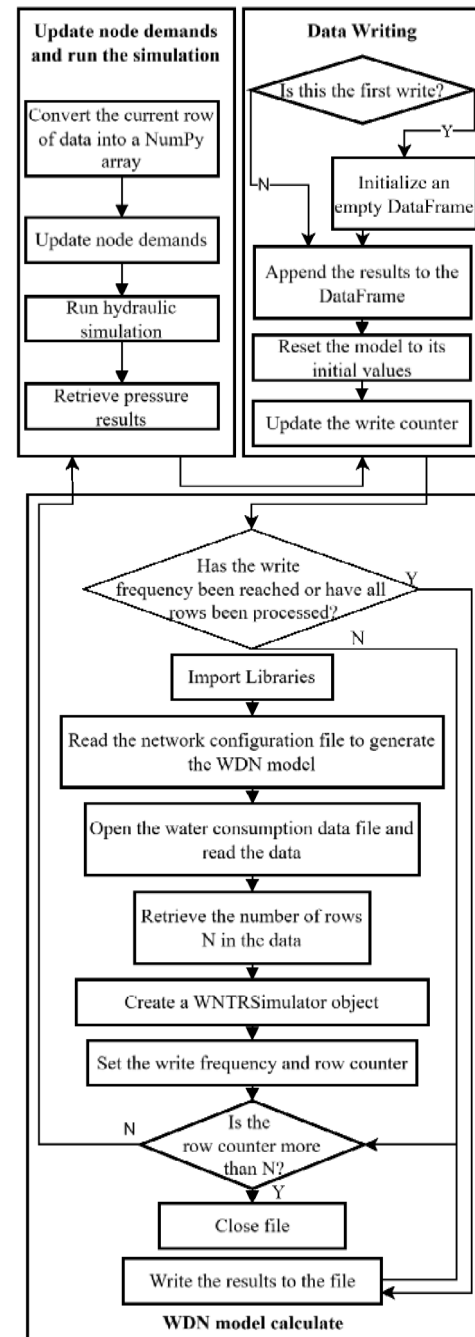


FIGURE 4 Hydraulic computation framework for simulated water distribution network.

leakage database, triple-leakage database, and quadruple-leakage database. Taking the single-leakage database as an example, 10,000 samples were randomly selected from the 14,400 single-time-snapshot hydraulic state samples under normal conditions, and 2500 samples were randomly selected from each of the four specific leakage scenarios under single-leakage conditions, totaling 10,000 samples. These 20,000 samples collectively constitute the single-leakage database.

**TABLE 1** Convergence behavior across scenarios.

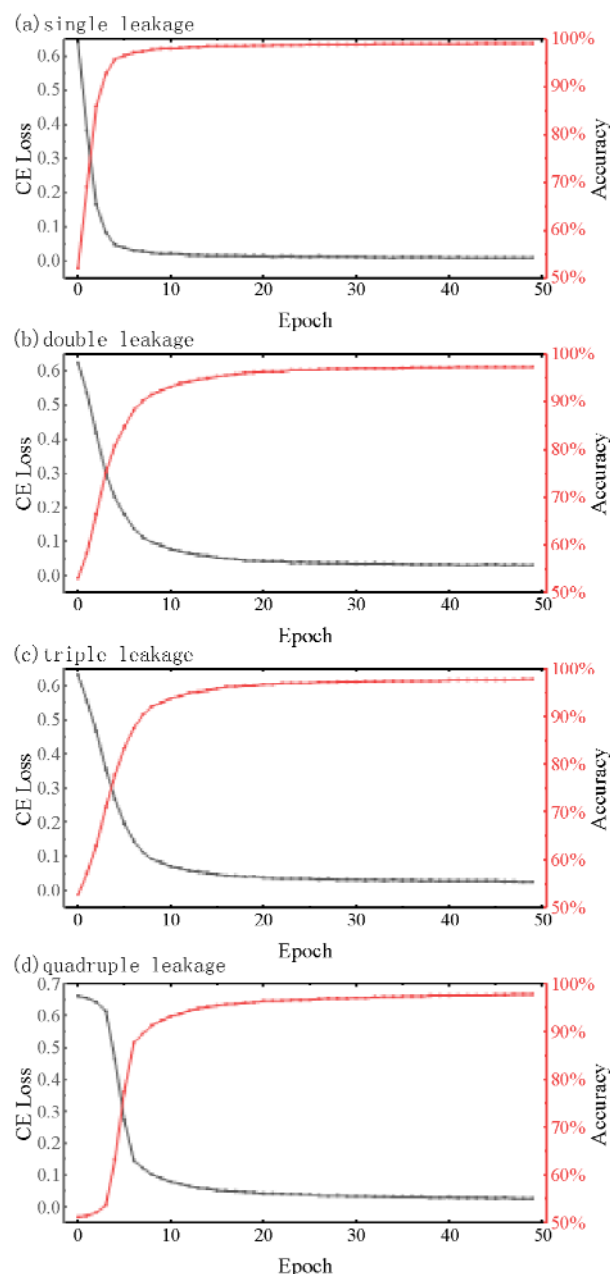
Scenario	Rapid	Inflection	Steady
Single	1–6	7–15	16–50
Double	1–8	9–20	21–50
Triple	1–7	8–18	19–50
Quadruple	1–7	8–17	18–50

3.1.2 | Discussion on the full-feature model

First, the simulated WDN data (node pressure and node water consumption), network topology data, and network state information are input into the graph database as the node-feature matrix, adjacency matrix, and graph labels, respectively. These inputs form four databases: the single-leakage database, the double-leakage database, the triple-leakage database, and the quadruple-leakage database. A horizontal comparison among these combination is conducted to determine whether different combinations of leakages affect the recognition of the overall graph state. During the training process, the CE loss function is used to measure the difference between the predicted results and the true labels. A larger CE loss value indicates a greater difference between the predicted results and the true labels, while a smaller value signifies that the predictions are closer to the true labels. In WDN modeling, the CE loss not only quantifies the model's deviation in leakage detection but also indicates whether the GCN has effectively captured the associations between the topology and node attributes. A smooth decline in the convergence curve often indicates that the model is gradually learning feature representations that best differentiate the states of the water network, providing a reliable basis for leakage detection and localization. The convergence comparison of those is shown in Figure 5.

Figure 5 depicts the training convergence behavior of the model across four leakage scenarios—single, double, triple, and quadruple leakage—with progressively increasing complexity. Across all scenarios, cross-entropy loss (CE loss) and accuracy (ACC) follow a clear three-stage trajectory comprising a rapid-decrease stage, an inflection stage, and a steady stage. The epoch ranges associated with each stage for the respective scenarios are summarized in Table 1.

In the rapid-decrease phase of the single-leakage scenario, the loss drops from 0.6435 to 0.393, indicating effective initial self-adaptation; ultimately, the loss and ACC stabilize at approximately 0.0106 and 99.1%, respectively. For the double- and triple-leakage scenarios, the initial loss is lower (about 63%) yet decreases sharply, stabilizing at 0.0315 and 0.0259, with ACC reaching 97.4% and 97.8%. In the quadruple-leakage scenario, despite the highest initial

**FIGURE 5** Comparison of convergence speeds across different cases.

loss, substantial improvement is observed—particularly mid-training—culminating in stabilization around 0.026. As shown in Figure 5, the single-pipe leakage exhibits the fastest convergence among all scenarios, reflecting superior initial adaptability and yielding the highest final ACC. The quadruple-leak case has the highest initial loss. Over Epochs 1–4, the loss reduction is about 0.05592 for the single-leak case but only about 0.047 for the quadruple-leak case, with ACC rising to 92.8% and 53.7%, respectively. During Epochs 4–15, the quadruple-leak case shows the fastest loss decrease, suggesting rapid learning once complex leakage patterns have been accommodated.



In summary, the model exhibits the fastest convergence speed and the lowest final CE loss value in the single-leakage condition, indicating that the model can quickly and accurately learn single leakage types. The convergence speed and final CE loss values in the double-leakage and triple-leakage conditions are similar. The model can effectively learn multiple leakage types, though the convergence speed is slightly slower than in the single-pipe leakage condition. The quadruple-leakage condition shows a slower initial convergence speed but accelerates during the middle stage. Its final CE loss value is comparable to that of the double-leakage and triple-leakage conditions. Increasing the number of leakage mainly delays the initial convergence yet does not compromise the eventual performance: across all scenarios, the final ACC exceeds 97%, the CE loss remains within 0.01–0.03, and the average confidence reaches 97%–98%. Taken together, these outcomes indicate that the proposed GCN captures the nonlinear coupling associated with multiple leaks and delivers consistently high accuracy and robustness across scenarios.

After the detection of leakage conditions in the WDN, localization of the leaking pipe is performed. This localization is based on the node importance scores obtained during the leakage condition detection process. Figure 6a illustrates the risk contour map for single leakage.

The topologies formed by all monitoring layouts and the network exhibit rotational and axial symmetries: with nine, four, or one monitoring point(s), the configuration is 90° rotationally symmetric, whereas with two monitoring points it is 180° rotationally symmetric. Taking the single-leakage case as an example, a leak on pipes 3–4 is analogous to leaks on pipes 3–5, 5–7, 7–8, 8–11, 4–9, 9–10, and 10–11; the primary difference is that, when leakage occurs on these respective segments, the risk indices at other nodes become comparatively higher. Accordingly, the selected leaking segments can serve as representative surrogates for the majority of leakage occurrences in the network.

The WDN risk index is defined by the method provided in Section 2.5.2. The risk level of each pipeline can be clearly observed in the Figure 6. The higher the risk level of a pipeline, the greater the probability of leakage occurrence. In each graph classified as abnormal, the pipeline segment with the highest risk level is identified as the leaking segment. Leakage occurring in different pipes can be easily observed in the risk contour maps. Figure 6a1 shows leakage in pipes 6–10; Figure 6a2 shows leakage in pipes 3–4; Figure 6a3 shows leakage in pipes 7–8; and Figure 6a4 shows leakage in pipes 8–11. T of the single-leakage network is 99.53%, and T' is 99.95%.

Double leakage is categorized into two types (b and c), where type b represents leakage in two non-adjacent

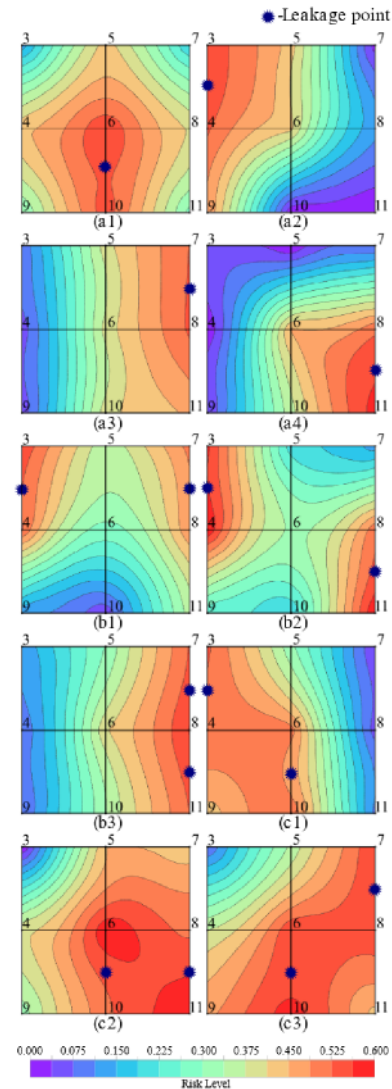


FIGURE 6 Risk contour map of simulated water distribution network.

pipe endpoints, and type c represents leakage in two adjacent pipe endpoints (where two endpoints of the leaking pipes are connected by a non-leaking pipe). When double leakage is of type b, the leaking network can be clearly identified (Figure 6b). As shown in Figure 6b1, the leaking pipes are 3–4 and 7–8; in Figure 6b2, the leaking pipes are 3–4 and 8–11; and in Figure 6b3, the leaking pipes are 7–8 and 8–11. In contrast, type c leakage cannot be directly determined from the leakage risk (Figure 6c). As observed in Figure 6c1, the conclusion can be drawn that pipes 3–4, 4–6, and 6–10 all exhibit leakage, which leads to a false positive for pipes 4–6. Since the weighted leakage risk of pipes 4–6 (the average leakage risk of all points on the pipe) is higher than that of pipes 3–4 and 6–10, it may result in false negatives for segments 3–4 and 6–10. The cause of this error is that there are no additional nodes between nodes 4 and 6, so their labels can only be obtained

TABLE 2 Leakage localization accuracy.

Leakage type	$T(\%)$	$T'(\%)$
Single leakage	99.53	99.95
Double leakage	94.81	98.56
Triple leakage	86.38	95.47
Quadruple leakage	83.63	93.26

through interpolation, and their true hydraulic states cannot be captured. T of double leakage is 94.81%, and T' is 98.56%. The situations for triple leakage and quadruple leakage are similar to those for double-pipe leakage, and the localization performance is given by Table 2.

3.1.3 | Discussion of the number of monitoring points

It is practically impossible to install demand meters and pressure gauges on all nodes in a WDN. In reality, only a subset of nodes in a WDN is equipped with pressure gauges. These pressure gauges are typically installed at key nodes, such as water sources and pump outlets, branch points of main pipelines, critical user nodes, network end-points, and high or low elevation areas. This necessitates that the model can achieve results with minimal deviation without requiring pressure and water consumption data for all nodes. The following section will present a study on reducing the proportion of monitoring points used in the GCN.

Since the feature vector of each node in the GCN is two dimensional (2D), consisting of node pressure and node water consumption, the two will be discussed together. Node water consumption data is easier to obtain compared to node pressure data, as human water demand does not change due to pipeline network incidents. Under the same external conditions, human water demand remains relatively constant. Current research has shown that water consumption can be predicted with a maximum relative error below 20% by incorporating factors such as daily weather conditions (Li et al., 2023). Therefore, in subsequent research, estimated water consumption data (predicted water consumption) with a maximum relative error of 20% will be used to replace the precise water consumption data in the full-feature model.

The features input into the model were as follows: nine precise pressure monitoring points and four predicted water consumption points (Figure 7a); four precise pressure monitoring points and four predicted water consumption points (Figure 7b); two precise pressure monitoring points and four predicted water consumption points (Figure 7c); one precise pressure monitoring point and four predicted water consumption points (Figure 7d). In

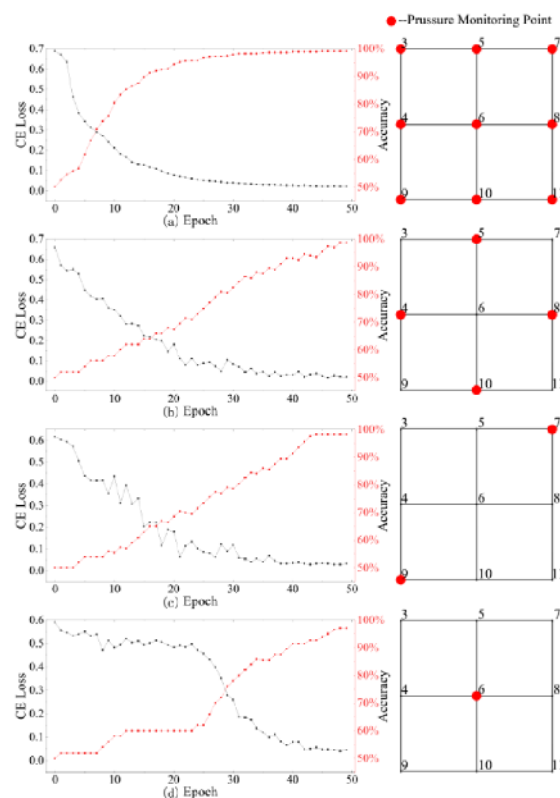


FIGURE 7 Convergence curves and arrangement of pressure monitoring points.

these four scenarios, the proportions of pressure monitoring points were set as 100%, 44%, 22%, and 11%, respectively. The arrangement of these pressure monitoring points and convergence curves is illustrated in Figure 7.

Four groups of data were observed, and it was found that the reduction in the number of monitoring points impacts the model's convergence. However, a single monitoring point is still sufficient to support effective model convergence.

In Figure 7a, the configuration with nine pressure sensors performs best: the loss decreases smoothly from 0.6869 to a final value of 0.0216, while ACC rises from 50% to 99.1%; the curves are very smooth, indicating strong convergence. By contrast, in Figure 7b (four pressure sensors), the loss curve exhibits mild mid-course fluctuations but maintains a clear downward trajectory, converging to 0.0214 approximately; ACC reaches 98.7%, slightly lower than Figure 7a yet still indicative of stable performance. In Figure 7c, when only two pressure sensors are available, the CE loss exhibits more pronounced fluctuations. Starting from 0.6158, the CE loss oscillates frequently during descent yet still converges successfully to 0.03289. The ACC reaches 98.3%. Notably, the CE loss fluctuates markedly over Epochs 8–23, whereas ACC varies only slightly, indicating that classification confidence is

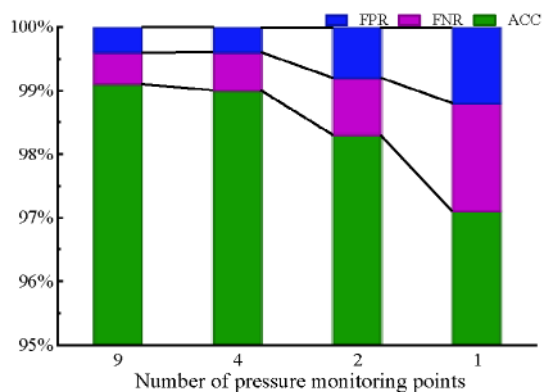


FIGURE 8 Performance metrics of pipe leakage detection.

primarily being improved during this interval. Overall, although convergence with 22% sensor coverage is less smooth than in the first two settings, the results remain satisfactory. In Figure 7d (one pressure sensor), a noteworthy phenomenon is observed: during Epochs 1–23, the CE loss decreases only from 0.5897 to 0.4969 and shows pronounced oscillations. ACC, however, either remains flat or increases, with no declines observed. This suggests that, during ACC plateaus, prior class assignments are retained while confidence is prioritized, with adjustments to sample classifications made afterward. After Epoch 24, key features are captured and convergence accelerates. This could be attributed to the presence of only one pressure monitoring point, as the remaining demand monitoring points cannot provide direct pressure information. Since the GCN model relies on adjacency relationships in the graph structure for information propagation, the use of a single pressure monitoring point limits the extent to which information can spread in the graph. Consequently, the information from this node can only be effectively transmitted to neighboring nodes, while non-monitored nodes and demand monitoring points fail to supplement sufficient pressure data. This directly impacts the model's overall predictive ability for node pressures, causing the gradient descent speed to gradually slow down. Nevertheless, after 50 epochs, the final CE loss value decreases to approximately 0.0452. The ACC reaches 97.1%. Although its performance is slightly inferior to conditions with multiple pressure monitoring points, it still converges effectively and supports the model's stability.

Figure 8 presents the algorithm's ACC, FNR, and FPR under different numbers of pressure monitoring points. It is visually evident that as the number of pressure sensors decreases, ACC drops from 99.1% to 97.1%, while FPR and FNR increase from 0.4% and 0.5% to 1.2% and 1.7%, respectively. Consistent with intuition, the decline in ACC and the rise in FNR and FPR are directly related to reduced sensing capability in the network.

Moreover, Figure 8 indicates an accelerated decline in ACC as sensor density decreases: relative to the first setting, the second has 56% fewer sensors with a 0.1% drop in ACC; the third has 22% fewer than the second with a 0.7% ACC drop; and the fourth has 11% fewer than the third with a 1.2% ACC drop. Due to the scale of the experimental network, the minimum feasible sensor density is 11% (at least one pressure sensor). If the number of sensors were reduced further, the ACC would likely deteriorate more rapidly. Additionally, when sensor density decreases, FNR (classifying leakage “1” as normal “0”) increases faster than FPR (classifying normal “0” as leakage “1”). This occurs because fewer sensors capture fewer manifestations of pressure drops. When leakage occurs, insufficient evidence is captured to corroborate the event, making the model more inclined to assigned leakage (label 1) as normal (label 0), thereby increasing FNR faster than FPR. Under the present simulation conditions, although convergence degrades as the number of pressure sensors decreases, a single pressure sensor still suffices for effective convergence, with the final loss remaining within an acceptable range.

3.1.4 | Discussion of algorithmic robustness

The pressure data in the simulation experiments are generated by a calibrated hydraulic simulator and contain no noise. Such ideal conditions are rarely encountered in practical applications. In engineering practice, typical industrial pressure sensors exhibit random measurement uncertainty on the order of ± 0.1 – 0.3 m (degrading to about ± 0.5 m in adverse scenarios). Accordingly, $\sigma = 0.3$ – 0.5 m can be regarded as a representative noise level in WDN. Therefore, to further assess robustness under imperfect calibration, information scarcity, and measurement uncertainty, this subsection examines the behavior of the trained classifier when pressure inputs are corrupted with Gaussian noise at varying intensities ($\sigma = 0.3, 0.5, 1.0, 2.0, 4.0, 8.0$, and 16.0 m).

A model with a single pressure sensor is selected to reflect robustness to pressure uncertainty under low monitoring density. The resulting ACC, FNR, and FPR under the different Gaussian-noise settings are shown in Figure 9.

Figure 9 indicates that at $\sigma = 0$ and 0.3 m the accuracy remains at 97.1%, implying negligible impact on performance. At $\sigma = 0.5$ m, accuracy decreases to 96.5%, marking the onset of degradation. When σ reaches 1.0 m (exceeding the random-error range typical of most sensors) ACC falls below 95%. Further increases to $\sigma = 2.0$ m and $\sigma = 4.0$ m reduce accuracy to 88.5% and 73.5%, respectively.

In addition, a trend similar to that in Figure 8 also appears in Figure 9: as noise increases, the rise in FNR

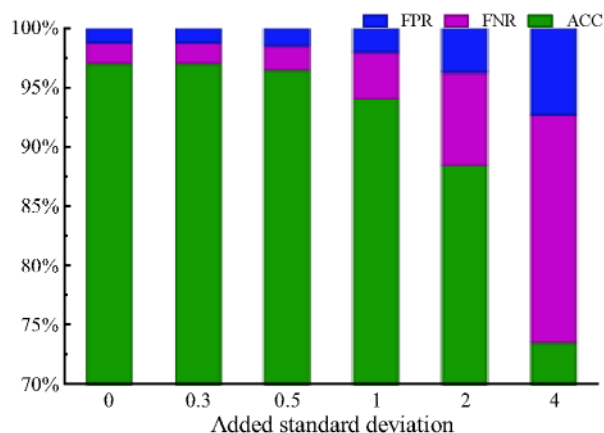


FIGURE 9 Performance metrics under different Gaussian noise.

(leakage classified as normal) is markedly faster than FPR (normally classified as leakage). This may be attributed to cross-entropy training, which constrains the predicted probability p to approach the true label y by minimizing the CE loss. When inputs are informative and class features are well separated, the model yields large positive-negative logit gaps, producing high-confidence predictions near 0 or 1. However, as input noise increases and the feature signal-to-noise ratio declines, the final-layer logits tend toward zero, causing outputs to cluster near 0.5. In combination with the asymmetric compression of negative logits by LeakyReLU, the model ultimately tends to classify samples as normal (0) in high-noise regimes.

Overall, within the attainable error range of pressure sensors in WDN, the model performance remains essentially stable; once noise exceeds 0.5–1.0 m, accuracy degrades and FNR increases markedly. The model’s performance on real WDN data (L-Town) will be discussed in Section 3.2.

3.2 | Experiments of WDN in L-Town

As discussed in Section 3.1 on small-scale simulated networks, GCN can be effectively coupled with WDN and used for abnormal state recognition and pipe burst region identification. In this chapter, GCN is integrated with the L-Town network to conduct modeling experiments on large-scale networks.

3.2.1 | Introduction of the L-Town WDN

L-Town is a benchmark model widely used in WDN research, initially introduced by BattLeDIM (Dataset of BattLeDIM: Battle of the Leakage Detection & Isola-

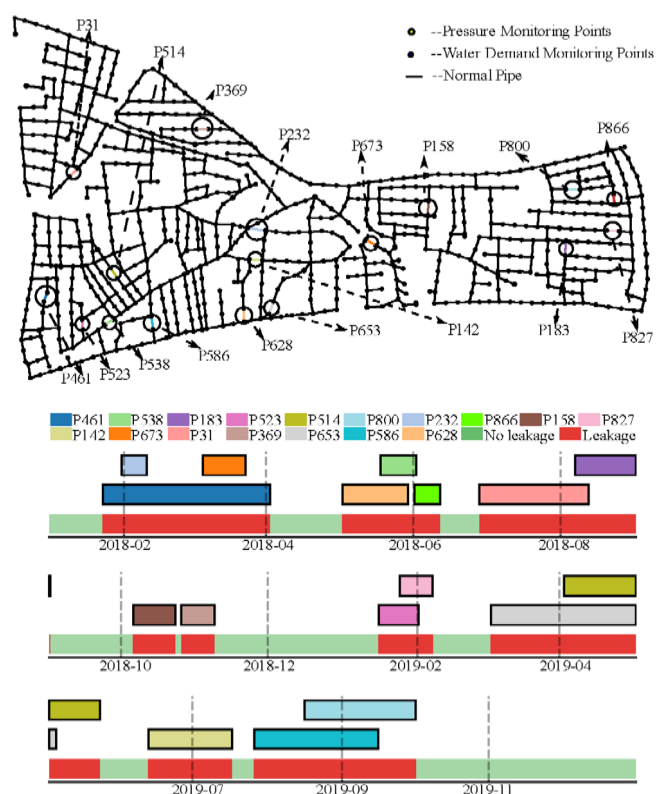


FIGURE 10 Full-scale water distribution network diagram.

tion Methods, 2020). The model provides a realistic yet anonymized WDN, along with extensive historical and evaluation datasets, offering valuable resources for researchers and engineers in the development and assessment of leakage detection and isolation methods.

The L-Town dataset includes detailed EPANET model files, comprising nominal parameters, actual network parameters, and historical baseline demands for nodes. In addition to the network model, the dataset provides extensive measurement data from the SCADA system, including: demand, including water consumption measurements from 82 automated meter reading devices; water-level data, including tank water-level measurements from 1 water-level sensor; pressure data, including pressure measurements from 33 pressure sensors.

In addition, the dataset includes detailed information and time-series data on leakage events, as well as reports on repaired leaks. These data provide a solid foundation for studying the operation of WDN, developing leakage detection algorithms, and evaluating system performance.

3.2.2 | Full-scale WDN model

This section first conducts anomaly-state recognition experiments on a full-scale WDN. Figure 10 illustrates the nodes and topology of the full-scale network in EPANET.



The Figure includes 33 pressure monitoring points (yellow dots), and real demand data are available for all nodes. According to the L-Town “readme” documentation: The nominal base demands for each node are based on average historical metered consumption. In general, the difference between the actual and the nominal values for each consumer type (residential and commercial) is less than 10%. Pressure data were obtained from pressure measurements from 33 pressure sensors in the L-Town network. These two types of real data provided by L-Town were used as the model inputs in this study.

In addition, Figure 10 presents the topological positions of leaking segments within the WDN together with the leakage occurrence times for each segment. The cumulative leakage duration amounts to 411.44 days, while the cumulative non-leakage duration is 318.56 days.

The overall approach for the abnormal state recognition experiment is the same as that in Section 3.1. First, the graph topology information in .inp format is exported from EPANET. Based on the endpoint data of pipes in the file, the network’s topology matrix is extracted. Next, the node information in the network is used as node vectors in the GCN to construct the node-feature matrix. Each node has a 2D feature, consisting of pressure and water demand. If a node has only water-demand information but no pressure information, the pressure is set to a default value of 0. The node-feature matrix, adjacency matrix, and network state labels together form the graph database.

The graph dataset was composed of two main categories: normal and leakage conditions. Among them, the normal condition, whose label was set as “0,” represented the hydraulic state data of the WDN when no leakage occurred in any pipe. The data were recorded at a 5-min resolution. From the 318.56 leakage-free days, a total of 300 days was randomly sampled, producing 86,400 graph instances labeled as normal (label = 0) that were stored in the graph database. The leakage condition, whose label was set as “1,” denotes the hydraulic-state data of the WDN when at least one of the 17 leaking pipe segments shown in Figure 10 is leaking (either a single segment or two segments concurrently). These data were likewise recorded at a 5-min resolution; from the 411.44 leakage days, a total duration of 300 days was randomly sampled, yielding 86,400 graph instances labeled as leakage (label = 1), which were stored in the graph database. In this way, a balanced graph database is obtained.

Through a two-layer GCN aggregation process over 50 epochs, the state classification of the network is output.

The initial CE loss value is 0.6482 and decreases rapidly to 0.4275 within the first 10 iterations. This plateau suggests possible issues like gradient vanishing, leading to insufficient parameter updates and suboptimal convergence.

To address this limitation, the Adam optimizer with a dynamic learning rate is introduced. As shown in Figure 10b, with Adam optimization, the CE loss continues to decrease, reaching 0.2722 at the 50th iteration—approximately 23% (percentage difference in final CE loss relative to the baseline method) lower than the original method’s final value, confirming the effectiveness of Adam in enhancing model convergence.

Although the optimization strategy improved the model’s convergence speed and reduced the CE loss value, the final convergence performance remains suboptimal. In L-Town, 33 pressure sensors are deployed across 782 nodes, yielding a sensor coverage of 4.2%. Under this configuration, the model achieves an ACC of 78.6%, while FPR and FNR increase to 5.9% and 15.5%, respectively. The faster rise of FNR relative to FPR mirrors the behavior observed in the validation network and suggests that, when feature extraction is limited, the classifier is biased toward labeling leakage states as normal. The main reason may be the sparsity of nodes with pressure information in the graph. The nodes with pressure information are far apart, preventing effective propagation and aggregation of information within the graph, thus limiting the model’s ability to learn global pressure features. As a result, the overall feature extraction of the model is insufficient, and the model fails to fully capture the features of key nodes, affecting its overall predictive performance. Due to the absence or weakening of critical information, the model struggles to find the global optimum during training, making it difficult to further reduce the CE loss value. This results in the difficulty of model convergence. Therefore, to further improve the model’s convergence performance, the network model needs to be simplified to enhance the ability of information transmission between distant nodes in the model structure.

3.2.3 | Modeling of the simplified WDN

The method of node clustering is introduced (Giustolisi & Ridolfi, 2014) based on the flow direction in pipes and the pressure characteristics of nodes, merging nodes similar to the pressure monitoring points. The connectivity between nodes is determined based on the flow direction information in the pipes. The pipes subject to leakage were replaced by areas prone to leakage. In this way, the topology of the WDN is simplified (Figure 11a). In terms of procedure, the construction of the overall samples was consistent with that described in Section 3.2.2. The difference was that, in the simplified WDN, the mean water demand of the area and the pressure at pressure monitoring points were used as node features. Of the dataset, 90%

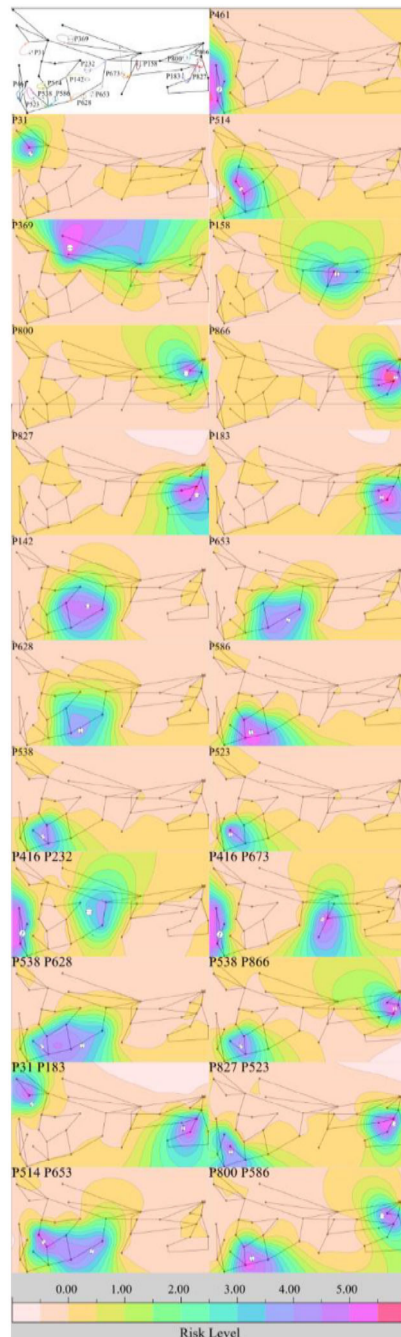


FIGURE 11 Simplified water distribution network diagram and risk level.

was used as the training set, and the remaining 10% was used as the test set.

The state of the WDN is identified using the GCN. The model's convergence significantly improves after the topology is simplified (Figure 11a). The CE loss during training shows a significant downward trend, demonstrating good convergence (Figure 12). Initially, the CE loss is 0.6854, and the model begins learning from the data. In subsequent iterations, the CE loss decreases rapidly, dropping to 0.1123 by the fifth iteration. This rapid decline indicates that the

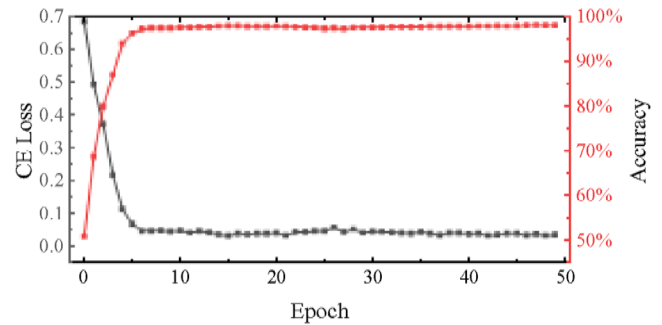


FIGURE 12 Convergence curve of the simplified water distribution network training.

model effectively captures the main features of the data at an early stage.

By the sixth iteration, the CE loss has decreased to 0.0673, a reduction of approximately 0.62 from the initial value, demonstrating the model's strong learning capability. Subsequently, the CE loss continues to decrease and fluctuates within a low percentage range, reaching 0.0425 by the tenth iteration. The model reduces the CE loss value to nearly 0.04 within a short time frame.

In the subsequent training process, the CE loss value fluctuates slightly between 0.032 and 0.04 and overall remains at a low level. These small oscillations may arise because, while maintaining classification accuracy, the algorithm further reinforces confidence via the CE loss, thereby manifesting as a flat ACC curve accompanied by persistent, low-amplitude fluctuations in CE loss.

Notably, the CE loss value falls below 1% in multiple iterations, such as 0.0335 at the 34th iteration and a minimum of 0.032 at the 49th iteration, further demonstrating the model's robustness and accuracy. The model's final output CE loss is 0.0347.

On the test set, the model achieves a prediction accuracy of 95.3% for the state of the WDN, with a FNR of 0.30% and a FPR of 0.17%.

The GCN output for the network graph with abnormal states is used to calculate node importance rates, obtaining the leakage-risk values for the nodes. Using the network topology as the base map, leakage contour lines are plotted on it, resulting in the risk contour map of the WDN (Figure 11). In each subfigure, the leaking pipe segment is annotated in the upper-left corner, and a short line segment circled in white indicates its location in the actual WDN. In the first subfigure, arrows sharing the same color as the pipe segments depict the mapping between the leaking segment and the monitoring point. Monitoring points corresponding to a given segment are colored identically to that segment. When a single monitoring point is associated with two segments, distinct fill and stroke colors are used to represent both associations.



It can be observed that the contour maps precisely pinpoint the areas with the highest leakage risk to the pressure monitoring points in the respective pipe zones, and the leakage locations can be distinctly identified. The performance evaluation system for localization in the corresponding zones was similar to that for pipe leakage described in Section 2.5.2, except that the pipes in the equations were replaced by the corresponding zones.

Furthermore, leakage segments are not naively localized at the pressure sensors. A leaking pipe jointly perturbs multiple nearby sensors, and the GCN in this method is expressly designed to exploit graph topology; consequently, the leakage risk at every monitoring point (the likelihood of a leakage within its coverage area) is produced by the algorithm. These risk values capture the association between pressure sensors and segments. Take subfigure P142 as an example: although P142 is clustered to the left sensor in the graph-simplification step, it actually lies between the coverage regions of the left and right sensors. Accordingly, elevated leakage risks are produced for both sensors. By combining risk values at other nodes, a leakage-risk contour map can be constructed. By analogy to geomorphology, a “ridge” is formed by connecting outward-bulging vertices of contours; if these ridges emanate from a single peak, their intersections indicate the peak’s location. Figure 11 shows four such prominent “ridges” in subfigure P142, whose intersection lies notably closer to the leaking segment than to any sensor. The same pattern appears in subfigures P653, P628, P538, and others. Therefore, the localization accuracy of the method can be regarded as between district-level and pipe-level granularity.

Moreover, although P586 and P538 are clustered to the same sensor, their risk-contour maps differ: the topmost concentric contour appears above the sensor for P538 and to the right for P586, in both cases covering the leakage area. This indirectly corroborates that the proposed localization mechanism outperforms node-centric multiclass methods that first cluster the network and then perform classification.

Naturally, several limitations of the proposed localization algorithm were identified in this study:

1. When a leakage exerts a very strong influence on only one pressure sensor while affecting the others similarly, the localization accuracy approaches that of district-level localization; this is observed in subfigures P31 and P183.
2. When two leakage simultaneously, the estimated location may be biased toward the segment between them; this occurs in subfigures P514 and P653, and the error mechanism has been discussed in Section 3.1.2 with reference to Figure 6c.
3. False negatives may also arise (subfigure P514 P653): P514 involves an abrupt leak and is closer to two sensors, thus producing a stronger impact; P653 involves an incipient leak and lies farther from both sensors, thus producing a weaker impact.

3.2.4 | Comparison with similar algorithms

L-Town is a widely used benchmark model in WDN research, originally introduced by BattLeDIM. Accordingly, numerous studies have evaluated leakage-detection algorithms on this benchmark. Daniel et al. (2022) employed a two-stage framework combining sequential detection and model-based localization; a semi-supervised 1D-CNN autoencoder was used to reconstruct pressure signals for leak localization (Tornyeviadzi & Seidu, 2023); a graph-based state interpolation (GSI) was performed to complete hydraulic states and applied dictionary-learning (DL) classification, achieving localization from pressure series sampled at 5-min intervals (Irofti et al., 2022); Ma et al. (2022) combined SCADA-driven state estimation with real-time demand calibration and history-simulation differencing for leakage detection; Wan et al. (2023) used an Artificial Neural Network (ANN) for multi-step forecasting of window-level future flows, employing cosine-distance residuals with EWMA smoothing to suppress noise and set leakage-alarm thresholds.

Analytical methods also constitute a mainstream family of approaches for leakage identification. As early as 2016, Delgado-Aguinaga et al. (2016) proposed multi-leak diagnosis in pipelines based on extended Kalman filter; the core of such methods is to analyze sustained, high-resolution pressure series and detect leaks by contrasting features between leakage and normal states. The driving paradigm of these methods is not aligned with that used in the present approach. Because the two study datasets employed here (the simulated WDN and L-Town) do not contain the high-resolution pressure series required by those methods, an effective head-to-head comparison could not be performed. A coarse, mechanism-level comparison is still possible: only a sampling rate of 100 Hz is reported, and for typical problem sizes the computational cost is on the order of 5–80 Million Floating-point Operations Per second (MFLOPs). In this study, under the binary classification setting ($c = 1$), the two-layer GCN has a single forward-pass computational complexity of $O(nfh + eh + nhc + ec + nh + nc)$. Converting each multiply-accumulate into 2 Floating-point Operations Per second (FLOPs) and substituting the graph size used in this work ($n = 34$, $e = 44$, $f = 2$, and $h = 64$), the single-sample inference cost is approximately 0.0232 MFLOPs. Considering a constant factor for backpropagation and parameter



updates of $\alpha \approx 3$, one training step per graph is approximately 0.0696 FLOPs. Although longer time is consumed during training, inference for a single sample is at the millisecond level once the model has been trained. The referenced work does not analyze noise robustness; its sensor data indicate a maximum relative pressure error of 5%, and, if 10% Gaussian noise were also injected, its precision would likely fall below that of the present algorithm.

It should be emphasized that all six localization methods are primarily driven by long time-series hydraulic features, thereby enabling pipe-level precise localization.

In contrast to the above driving paradigm, supervised CNN and MLP models (Basnet et al., 2023) were employed for leak localization in L-Town. These studies systematically examined scenarios of increasing complexity—no noise, demand noise, mixed noise, and random leak—to localize the leakage district in L-Town. The findings indicate that leakage in WDN can be effectively identified when given the close similarity of their modeling paradigm and research objective to those of the present work, subsequent comparisons in this section were focused on these two models (CNN and MLP). Multi-source inputs (pressure plus demand) are used.

Robustness analyses for CNN and MLP were performed by adding Gaussian noise to demand and pressure data. Their no-noise setting corresponds to the single-sensor, full-feature simulated network used in this study. The demand-noise setting adds 10% Gaussian noise to demands while retaining simulated, noise-free pressures. This is analogous to Section 3.1.3 in which demand data with 20% Gaussian noise and exact pressures were used as inputs. Accordingly, outputs from the single-sensor model in Section 3.1.3 were used for comparison with the demand-noise case. The mixed-noise setting further adds 10% Gaussian noise to post-simulation pressure and demand under both leakage and non-leakage, resembling real measurements. Therefore, results from Section 3.2.3 were used for comparison with mixed noise. Additionally, Basnet et al. configured different thresholds for their models: 0.1 being the strictest and 0.9 the most lenient. Hence, in the comparative analysis the most lenient threshold (0.9), which yields the best performance for CNN and MLP, was adopted.

The resulting ACC, FNR, and FPR are tabulated in Table 3 and visualized in Figure 13.

By combining Table 3 with Figure 13, the following conclusions can be drawn:

Under the most lenient decision threshold for CNN and MLP, both CNN and GCN perform well in the no-noise scenario; CNN holds a slight edge, GCN is comparable, and MLP lags behind.

With demand noise, CNN maintains high precision and recall with only gradual declines; GCN shows relatively stable precision but a more pronounced drop in recall, reflecting a more conservative detection tendency; MLP

TABLE 3 Classification metrics of different models.

Noise level	Metrics	MLP	CNN	GCN
No noise	P	98.00%	99.20%	99.16%
No noise	R	96.80%	98.90%	98.84%
No noise	F1	97.40%	99.00%	99.10%
Demand noise	P	96.00%	99.00%	99.00%
Demand noise	R	93.40%	98.30%	96.60%
Demand noise	F1	94.70%	98.60%	97.09%
Mixed noise	P	87.10%	93.40%	96.51%
Mixed noise	R	80.40%	89.50%	94.00%
Mixed noise	F1	83.60%	91.40%	95.24%

Abbreviations: F1, F1 score; P, precision; R, recall.

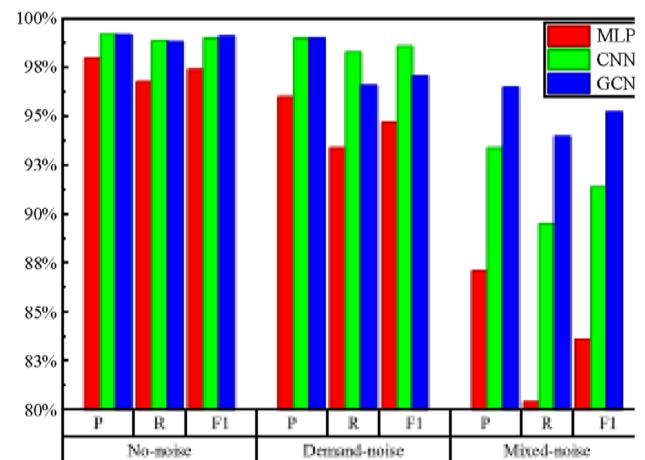


FIGURE 13 Classification metrics of different models.

degrades across all three metrics and at a faster rate. This indicates that, compared with CNN, the present method relies more on a combined pressure–demand signature for monitoring; a nearly unchanged precision (P) alongside declines in recall (R) and F1 implies reduced sensitivity to the leakage class (label = 1) and thus more false negatives.

Under mixed-noise conditions, GCN is the most stable with minimal degradation, CNN shows moderate performance with some deterioration, and MLP degrades the most and is most sensitive to noise.

Overall, accuracy in the absence of noise is led by CNN and GCN (with CNN slightly ahead); robustness under complex perturbations ranks as $GCN > CNN > MLP$; in terms of degradation rate, MLP deteriorates fastest as noise grows, CNN remains relatively steady, and GCN shows the strongest resistance to compound noise.

These differences can be attributed to inductive biases: GCN's topological aggregation preserves spatial consistency under compound disturbances; CNN's temporal convolutions smooth measurement noise; MLP lacks explicit spatiotemporal structure and is therefore more sensitive to uncertainty.



Because specific values of kernel size, stride, and padding for each CNN layer were not available. The referenced paper reports only the number of convolutional layers. The kernel lengths cannot be determined; in addition, details on pooling and the dimensionality after flattening are missing, so a quantitative compute comparison with the present algorithm cannot be made. Nevertheless, the binary setup in the present algorithm makes second-layer propagation and readout scale linearly with the number of classes, yielding clear efficiency advantages over multiclass configurations in both training and inference.

4 | CONCLUSIONS

This study proposes a GCN-based method for detecting and localizing pipeline leakage in WDNs. By integrating WDN structure data and node features into a GCN model, accurate identification and localization of leakage regions are achieved.

Simulation experiments confirm the effectiveness of the GCN method, demonstrating high adaptability and accuracy across various leakage scenarios. Specifically, in single-leakage scenarios, the GCN model maintains efficient convergence, highlighting its suitability for leakage localization.

Tests on the large-scale L-Town network initially faced convergence difficulties due to sparse monitoring data. However, adopting the Adam optimization algorithm significantly improved convergence, reducing the final CE loss to 0.42%, achieving a prediction accuracy of 99.65%, with an FNR of only 0.35% and no false positives. Leakage contour mapping further verified the method's capability for precise leakage localization.

Additionally, experiments indicated that while reducing monitoring points affects model convergence, the model maintains acceptable accuracy levels with strategically placed key sensors, providing valuable insights for resource optimization.

In addition, certain limitations were acknowledged in this study. Specifically, a GCN framework was constructed using 2D node features (pressure and demand) and the feasibility of leakage detection and localization under sparse monitoring was validated. Given the complexity and heterogeneity of WDN operation, future work may extend node features from two to multiple dimensions, without altering the framework, to enhance the separability of subtle leaks and robustness across operating conditions. For example, node elevation affects pressure but does not vary with the presence or absence of leakage. Introducing elevation as a feature allows coupling with pressure so that relative, rather than absolute, pressure changes under leakage are better captured. Incorporating temporal features is

likewise likely to be beneficial. Hour-of-day, day-of-week, and season strongly influence demand. Including such features can sharpen discrimination between consumption and leakage. Pipe physical attributes (diameter, roughness, length, material, and service age) can be added as edge features to improve aggregation of node information.

Furthermore, the current framework can be extended to spatiotemporal graph models (e.g., temporal GCN) to characterize dynamic propagation of hydraulic information and thereby improve early-leak detectability under evolving operating conditions.

Beyond the proposed GCN framework, several recently developed learning paradigms could further enhance leak detectability and localization in complex and dynamic water distribution systems.

Neural dynamic classification (Rafiei & Adeli, 2017b) maximizes the inter-class margins while discovering the optimal subset of discriminative features through a robust neural dynamic optimization. This mechanism could help distinguish small and weak leaks—characterized by low disturbance amplitudes easily masked by normal consumption fluctuations—from normal operating states, thus reducing false negatives (FNR). A limitation of the present study is that, due to its inherent mechanism, the GCN framework tends to classify marginal leakage conditions as normal events (as discussed in Section 3.1.4). By enlarging the inter-class margin, the Neural Dynamic Classification (NDC) approach could address this issue and improve the classification robustness of the proposed model.

Finite element machine (Pereira et al., 2020) performs parameter-free, locality-based learning with quadratic complexity, enabling rapid model updates when sensor layouts change or new district metered areas (DMAs) are added. Its efficiency and adaptability make it particularly suitable for the operational constraints of real-world water utilities and on-site deployments.

Dynamic ensemble learning (Alam et al., 2020) automatically balances accuracy and diversity among multiple neural submodels. Given the inherent heterogeneity and distributional shifts in WDNs—caused by daily demand variations, pump switching, or valve adjustments—the DEL mechanism can dynamically adapt to different operating regimes and pressure zones, thereby improving cross-condition stability and reducing output variance.

Finally, self-supervised learning (Rafiei et al., 2024) offers a promising solution for dealing with large volumes of unlabeled SCADA pressure and flow time-series data.

By pretraining on multi-time graph snapshots to learn structure-hydraulic coupled representations and then fine-tuning with limited labeled data, Self-Supervised Learning (SSL) can significantly enhance detection and localization performance under low-label and high-uncertainty conditions.



These methods collectively provide meaningful directions for improving the robustness, adaptability, and practicality of GCN-based leakage detection in future smart water networks.

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